

### SPL-3 Project Proposal Form, 2026

Institute of Information Technology (IIT)

University of Dhaka

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| <b>Student's Name:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Habibur Rahman Mahin</b> |               |                    |
| <b>Student's Roll:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>BSSE1422</b>             | <b>Phone:</b> | <b>01700808679</b> |
| <p align="center"><b>APEX-Net: Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localization</b></p> <p><b>Project Description:</b></p> <p>APEX-Net is a web application that lets anyone upload a chest X-ray and instantly receive an AI-powered analysis across 14 thoracic conditions. Alongside each diagnosis, the app displays a visual heatmap highlighting the exact regions of concern - making results easy to understand without a medical background. APEX-Net is designed to be publicly accessible, reliable across both common and rare conditions, and accurate enough to serve as a meaningful first-pass screening tool.</p> <p><b>Problem Statement &amp; Objectives:</b></p> <p>Problem Statement</p> <p>Thoracic diseases are among the leading causes of death worldwide, yet access to skilled radiologists remains severely limited in underserved regions. APEX-Net addresses this gap by providing an automated, AI-powered chest X-ray analysis tool that is accurate, interpretable, and deployable in low-resource settings. Existing deep learning approaches fall short in three key areas: they struggle to detect rare but serious conditions due to skewed training data, they lack the anatomical reasoning that radiologists naturally apply, and they offer no visual explanation of where abnormalities were found - reducing clinical trust.</p> <p>Objectives</p> <ol style="list-style-type: none"><li>1. Build an AI model that focuses on anatomically relevant regions of the chest X-ray, rather than treating the entire image equally.</li><li>2. Ensure reliable detection of rare but serious conditions alongside common ones.</li><li>3. Tune detection sensitivity per disease for clinically meaningful accuracy across all 14 conditions.</li><li>4. Deliver predictions with visual heatmap overlays so clinicians can see exactly where the abnormality was detected.</li></ol> |                             |               |                    |

5. Package the system as a lightweight web application suitable for deployment in low-resource clinical environments.

**Proposed Solution:**

APEX-Net integrates three complementary components:

- **Smart Anatomical Focus** - The model learns where each disease typically appears in the chest and focuses its analysis on those regions, mimicking how a radiologist reads an X-ray.
- **Rare Disease Detection** - A custom training approach ensures the model reliably catches rare but serious conditions, not just the common ones that dominate most datasets.
- **Clinical Reliability** - Rigorous training techniques and per-disease tuning ensure the model performs consistently across real-world X-rays and all 14 conditions.

**References:**

- [1] Strick, D., Padilla, C.G., Huang, A., & Gardos, T. (2026). "An Open-Source Reproduction and Enhancement of CheXNet for Chest X-ray Disease Classification." arXiv preprint. <https://arxiv.org/abs/2505.06646>
- [2] Hanif, M.S., Bilal, M., Alsagga, A.H., & Al-Sagga, U.M. (2025). "Enhancing Multi-Label Chest X-Ray Classification Using an Improved Ranking Loss." Bioengineering, 12(6), 593. <https://www.mdpi.com/2306-5354/12/6/593>
- [3] Hossain, M.I., Zunaed, M., Ahmed, M.K., Hossain, S.M.J., Hasan, A., & Hasan, T. (2024). "ThoraX-PriorNet: A Novel Attention-Based Architecture Using Anatomical Prior Probability Maps for Thoracic Disease Classification." IEEE Access, 12, 3252-3269. <https://ieeexplore.ieee.org/document/10371304>

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**Signature of the supervisor:**

**Date:** 14th May, 2026

**Proposal Presentation Feedback:**

**Midterm Presentation Feedback:**